ASSIGNMENT 2. IDENTIFY YOUR SYSTEM & NETWORK IDENTITY

- **Current logged in user**

It is a person who is currently using a computer system at a given time.

- **Computer hostname**

It is a name of a computer that can be identified on a network. It helps to identify a device so other system can communicate with it.

- **IPv4 address**

It is a unique number attached to every device on a network which help with identifying and communicating with other IoT devices.

- **MAC address**

This is the physical address of every particular device . it is the identification of every device on the web.

- **Default gateway**

this is the medium between the internet and the local network. It sends packet from the user to the server and response from the server back to the user.

- **DNS server**

This is simply a translator that changes a name of a website to an IP address

Questions

1. Which info would an attacker find useful?
2. Why is MAC address important in a network

1. Ipv4 address

   Username/hostname

   DNS servers

2. It is unique identification of any IoT devices so it helps to facilitate coneections

ASSIGNMENT 3 — NETWORK CONNECTIVITY & TROUBLESHOOTING

PING

it is a command line that does two particular things, checking the connection of a website and measuring the response time

TRACERT

It shows a particular path taken to reach a destination or a website

PATHPING

The combination of both ping and tracert

While ping checks the availability and time taken to connect. Tracert shows the part taken to reach the final destination and pathpings does all of it together

Number of hops to Google 11

Why attacker uses tracert before launching an attack

To identify weak point

To know how data travel

To understand network structure

ASSIGNMENT 4 — OPEN PORTS & ACTIVE CONNECTIONS

- **Listening ports**

0.0.0.0:135
0.0.0.0:445
0.0.0.0:5040
0.0.0.0:5371

- **Established connections**

TCP 10.125.222.178.49547
TCP 10.125.222.178.75076
TCP 10.125.222.178.40752
TCP 10.125.222.178.64604

- **Process ID (PID)**

5064
7577
5609
4678

- **Application using port 80/443**

Service host- port 80

Brave –port 443

Questions :Why are open ports dangerous

It can be used as an entry point by an attacker

ASSIGNMENT 5: WINDOWS CMD

Questions

- Difference between absolute and relative paths?
- Why is navigation important for attackers?

	Absolute path	**Relative path**
Starts from	**C:**	**Current path**
Length	**Long**	**short**

IMPORTANCES OF NAVIGATION TO ATTACKERS

To avoid detection

To locate vulnerability

To find valuable target

Why malware use hidden /system attributes

To avoid been seen

Command used to create scheduled task-schtasks /create /tn "Testnotepad"/tr"notepad.exe"/sc once /st 14.31

Command used to delete the task-schtasks /query/tn"testnotepad/fo list/v

Command used to delete task
sdtrstes /quer tn "testnotepad"
/fo LIST /v

- /tn — transaction name
  /tr — transaction reference
  /sc — source code
  /st — status

Why would creating a task as system be dangerous if misused

Cause it could affect our system operation

To locate vulnerability
To find vulnerable target

g malwares use hidden system
allowed being seen
scheduled tasks

chtasks /Create /tn "TestNote"
otepad.exe" /sc once /st 19:31

arameters
/tn "TestNotepad"
/tr "notepad.exe"
/sc once
/st 19:31

Na
Prog
Run
Start

DNS - Server - This is what Chang
the name of a website to an IP- address
so Computers Can understand.

Info attackers will find useful

IPv4 address
Mac address
Username / Hostname
DNS Servers

Importance of Mac address - It is the un
identification of any IOT devices, so it
to facilitate Connections.

32. Ping - It is a Commandline tool does
particular things, Checking the Connect
of a Website and measuring the respon
time.
Tracert - It shows the path tak
reach a particular website from your

Pathping is the combination of pinging and tracert.

So while ping checks the availability and time taken to connect, tracert shows the path taken to reach the website you're trying to connect and pathping does all of it together.

Why might attackers use traceroute before launching attacks

- To identify weakpoint
- To know how to a trace
- To understand network structure

		address	Prot.	PID
4	Listening port -	0.0.0.0:135	TCP	1060
		0.0.0.0:445	TCP	4
		0.0.0.0:5040	TCP	3808
		0.0.0.0:5371	TCP	49